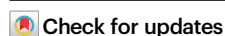


Demonstration of efficient predictive surrogates for large-scale quantum processors

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The ongoing development of quantum processors is driving breakthroughs in scientific discovery. Despite this progress, the formidable cost of fabricating large-scale quantum processors means they will remain rare for the foreseeable future, limiting their widespread application. To address this bottleneck, we introduce the concept of predictive surrogates, which are classical learning models designed to emulate the mean-value behavior of a given quantum processor with provably computational efficiency. In particular, we propose two predictive surrogates that can substantially reduce the need for quantum processor access in diverse practical scenarios. To demonstrate their potential in advancing digital quantum simulation, we use these surrogates to emulate a quantum processor with up to 42 programmable superconducting qubits, enabling efficient pre-training of variational quantum eigensolvers for families of transverse-field Ising models and identification of non-equilibrium Floquet symmetry-protected topological phases. Experimental results reveal that the predictive surrogates not only reduce measurement overhead by orders of magnitude, but can also surpass the performance of conventional, quantum-resource-intensive approaches. Collectively, these findings establish predictive surrogates as a practical pathway to broadening the impact of advanced quantum processors.

Modern quantum processors with hundreds of physical qubits^{1–3} have evidenced their scientific value across various domains, such as tackling synthetic sampling problems^{4–7}. At the same time, advancements in quantum error correction are steadily pushing the boundaries of fault-tolerant quantum computing^{8–12}. Despite the progress, the high cost of fabrication and maintenance hints that access to these advanced quantum computers will remain scarce for the foreseeable future. As a result, quantum-resource-efficient alternatives capable of emulating the behavior of large-scale quantum processors towards practical problems are highly demanded. A critical direction in this

pursuit is the development of effective and scalable methods to emulate the mean-value behavior of quantum processors, given its broad applications, including quantum physics^{13–16}, quantum chemistry^{17–19}, quantum system certification²⁰, and quantum machine learning^{21,22}.

Existing strategies to emulate the mean-value behavior of large-scale quantum processors can be broadly categorized into two classes: classical simulators and learning-based surrogates. Concrete examples of classical simulation encompass tensor-network^{23–26}, near-Clifford^{27–29}, and Pauli-path simulators^{30–33}. However, these classical

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simulators share a fundamental limitation, i.e., they are incapable of accurately and effectively capturing the intrinsic noise model specific to the quantum processor being explored. As a result, a non-negligible discrepancy may arise between the simulated results and the actual performance of the quantum processor. For learning-based surrogates, most of them adopt deep learning techniques to train neural networks that emulate the behavior of quantum processors^{34–37}. While these approaches have shown exceptional empirical results, they remain largely heuristic, lacking theoretical guarantees on their efficiency. Meanwhile, prior studies on learning-based surrogates with provable guarantees have primarily focused on noiseless or idealized quantum circuits, and their results cannot be straightforwardly extended to realistic noisy scenarios^{38–44}. To this end, a critical question arises:

Can we design provably efficient surrogates that emulate the mean-value behavior of quantum processors?

Here, we provide an affirmative answer to this question for estimating mean values of a quantum circuit $U(\mathbf{x})$, which consists of a gate fixed layout and d -dimensional input parameters $\mathbf{x}$. Our focus is on designing efficient predictive surrogates to emulate the mean value $\text{Tr}(\tilde{\rho}(\mathbf{x})\mathbf{O})$, where $\mathbf{O}$ is an observable and the noisy state $\tilde{\rho}(\mathbf{x})$ is generated by $U(\mathbf{x})$ on an assigned quantum processor. As shown in Fig. 1, given only limited access to the quantum processor, the optimized predictive surrogates can accurately estimate the mean values $\text{Tr}(\tilde{\rho}(\mathbf{x}')\mathbf{O})$ for any new input $\mathbf{x}'$ via purely classical inference. Note that many applications rely on mean-value estimation of this form. To provide concrete solutions, we devise two provably efficient predictive surrogates tailored to distinct scenarios, and experimentally exhibit

their efficacy in digital quantum simulation⁴⁵ using a quantum processor with up to 42 programmable superconducting qubits.

The first predictive surrogate, denoted by h_{cs} , extends the model from ref. 44 to the noisy case, where it serves as a classical shadow predictor⁴⁶. Specifically, h_{cs} is able to estimate mean values of $\tilde{\rho}(\mathbf{x})$ for many local observables under Pauli noise channels⁴⁷. Our theoretical analysis reveals that when $\mathbf{x}$ are independently and uniformly sampled from $[-\pi, \pi]^d$, h_{cs} is computationally efficient to achieve an ϵ -level prediction error, provided that either Pauli noise rates are relatively large or the gradient norm $\|\nabla_{\mathbf{x}} \text{Tr}(\tilde{\rho}(\mathbf{x})\mathbf{O})\|$ is upper bounded by a small constant. We experimentally assess the performance of h_{cs} in pre-training variational quantum eigensolvers (VQEs)⁴⁸ when applied to find the ground state energies of a class of 1D transverse-field Ising models (TFIMs). A notable feature of pre-training VQEs with the optimized h_{cs} is that it enables fully classical estimation of ground state energies for all TFIMs, substantially reducing measurement overhead relative to conventional VQEs. Experimental results indicate that the ground state energies estimated by h_{cs} are more accurate than those obtained from VQEs trained directly on the quantum processor, while using just 0.023% of the measurements.

The second predictive surrogate, denoted by h_{qs} , is designed to predict mean values $\text{Tr}(\tilde{\rho}(\mathbf{x})\mathbf{O})$ with $\mathbf{x}$ being correlated and a fixed observable $\mathbf{O}$. This setting is commonly encountered in the study of quantum simulation. Different from h_{cs} , h_{qs} supports the estimation of mean values for input parameters $\mathbf{x}$ sampled from arbitrary distributions. We prove that when the quantum processor is modeled by Pauli noise channels and $\mathbf{x}$ is sampled from a small range, h_{qs} is computationally efficient to reach an ϵ -level prediction error. We demonstrate

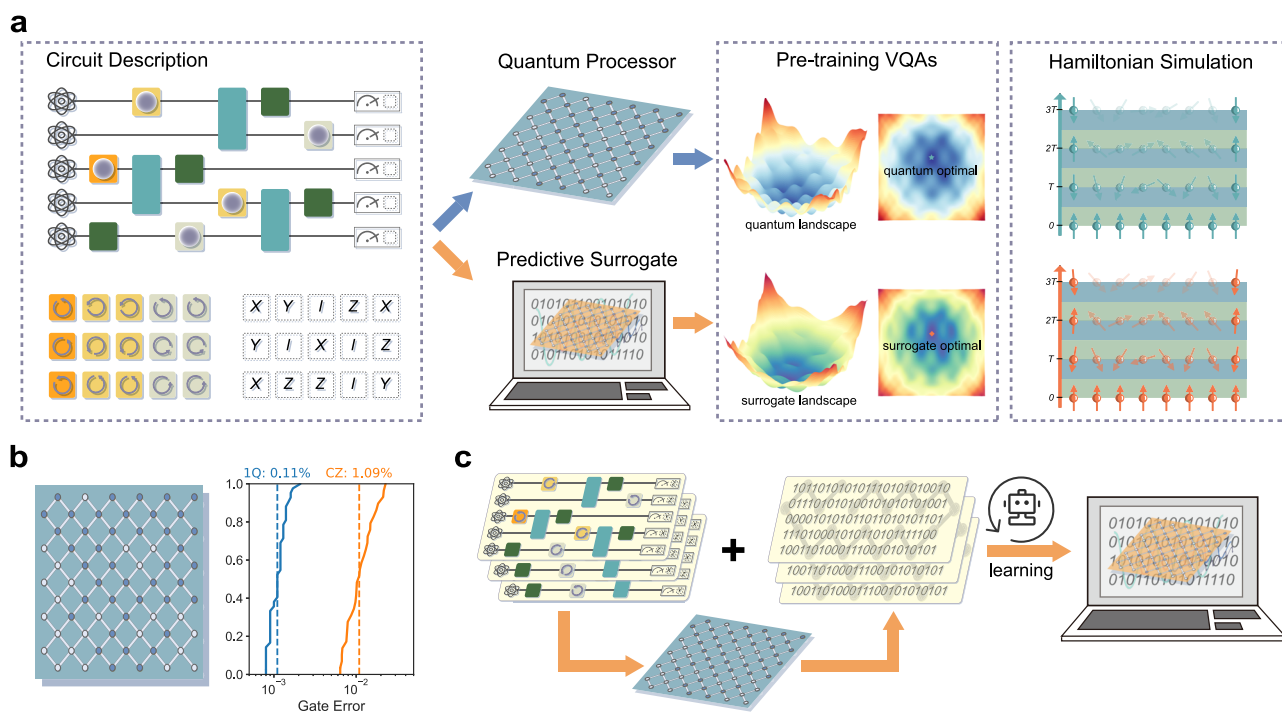


Fig. 1 | Framework for constructing predictive surrogates for quantum processors. **a** Given the classical description of a quantum circuit $U(\mathbf{x})$ with a fixed gate layout and tunable gate parameters $\mathbf{x} \in \mathcal{X}$, a predictive surrogate trained on data obtained from a quantum processor can emulate its mean-value behavior, i.e., $\text{Tr}(\tilde{\rho}(\mathbf{x})\mathbf{O})$, for parameters $\mathbf{x}$ spanning the entire space $\mathcal{X}$. Here, $\tilde{\rho}(\mathbf{x})$ represents the evolved state when executing $U(\mathbf{x})$ on a noisy quantum processor with a fixed initial state ρ_0 , and $\mathbf{O}$ refers to an observable. The optimized predictive surrogate can greatly facilitate tasks that typically require extensive access to the quantum processor, e.g., the pre-training of variational quantum algorithms and the characterization of phase diagrams in quantum many-body systems. **b** Overview of the employed quantum processor. The left panel illustrates the architecture of the

superconducting quantum processor used in this study. The blue circles highlight the employed subset of superconducting qubits in experiments, while the white circles represent the remaining idle qubits on the chip. The right panel indicates the cumulative probability distributions of Pauli noise errors for parallel single-qubit gates (blue) and parallel CZ gates (black) measured on the employed quantum processor. Besides, the dashed vertical lines denote the mean values. **c** The construction of the predictive surrogates consists of two steps. First, the classical learner collects the training dataset, i.e., labeled data obtained from quantum processors. Then, the classical learner leverages the trigonometric monomial expansion on the collected training dataset to implement the effective predictive surrogates.

the effectiveness and scalability of h_{qs} for pre-training VQEs on a 42-qubit quantum processor, and further showcase its applicability to identifying Floquet symmetry-protected phases, a distinct type of non-equilibrium states of matter⁴⁹. In contrast to the original approach⁵⁰, which demands extensive quantum processor access to estimate mean values across different $\mathbf{x}$ for phase identification, our method enables orders-of-magnitude reduction in measurement overhead by training a set of $\{h_{qs}\}$ to emulate mean-value behavior. Experimental results demonstrate that the optimized $\{h_{qs}\}$ can reliably probe the transition region of Floquet symmetry-protected phases of time-periodic Hamiltonians. These results open a new avenue for leveraging predictive surrogates to advance the study of digital quantum simulation.

Results

Problem setup

Let us begin by establishing the mathematical framework that characterizes the mean-value behavior of noisy quantum processors. Leveraging the universality of Clifford gates with RZ gates⁵¹, throughout this study, we focus on quantum circuits of the form $\mathbf{U}(\mathbf{x}) = \prod_{l=1}^d (\text{RZ}(\mathbf{x}_l) \mathbf{V}_l)$, where each Clifford operation $\mathbf{V}_l$ is composed of some Clifford gates (abbreviated as CI gates), and identity gates $\mathbb{I}_2$ with $\text{CI} = \{\mathbf{H}, \mathbf{S}, \text{CNOT}\}$.

To account for noise, the quantum circuit $\mathbf{U}(\mathbf{x})$ and its constituent gates are often described in terms of their quantum channel representations. Specifically, we denote the channel corresponding to $\mathbf{U}(\mathbf{x})$ as $\mathcal{U}(\mathbf{x})$, and the channels corresponding to the individual gates $\text{RZ}(\mathbf{x}_l)$ and $\mathbf{V}_l$ as $\mathcal{R}_Z(\mathbf{x}_l)$ and $\mathcal{V}_l$, respectively. Supported by the results of twirling operation⁵², we model $\mathcal{U}(\mathbf{x})$ as being affected by Pauli noise channels⁵³, i.e.,

$$\tilde{\mathcal{U}}(\mathbf{x}) = \bigcirc_{l=1}^d \tilde{\mathcal{R}}_Z(\mathbf{x}_l) \circ \tilde{\mathcal{V}}_l, \quad (1)$$

where $\tilde{\mathcal{R}}_Z(\mathbf{x}_l) = \mathcal{N}_p \circ \mathcal{R}_Z(\mathbf{x}_l)$, $\tilde{\mathcal{V}}_l = \mathcal{M} \circ \mathcal{V}_l$, and $\mathcal{N}_p$ and $\mathcal{M}$ denote the single-qubit and multi-qubit Pauli channels, respectively. Here $\mathcal{N}_p(\rho) = (1 - \bar{p})\rho + p_X \mathbf{X}\rho\mathbf{X} + p_Y \mathbf{Y}\rho\mathbf{Y} + p_Z \mathbf{Z}\rho\mathbf{Z}$ with p_X , p_Y , and p_Z being the noise parameters and $\bar{p} = p_X + p_Y + p_Z$. With these definitions, the mean-value behavior of a noisy quantum processor is characterized by

$$\mathbf{f}(\tilde{\rho}(\mathbf{x}), \mathbf{O}) = \text{Tr}(\tilde{\rho}(\mathbf{x})\mathbf{O}), \quad (2)$$

where $\mathbf{O}$ is a given observable and $\tilde{\rho}(\mathbf{x})$ denotes the quantum state ρ_0 evolved under the noisy circuit $\tilde{\mathcal{U}}(\mathbf{x})$.

The objective of the predictive surrogate is illustrated in Fig. 1a. That is, it is designed to learn a classical model $\mathbf{h}(\mathbf{x}, \mathbf{O})$ whose output approximates the true mean-value $\mathbf{f}(\tilde{\rho}(\mathbf{x}), \mathbf{O})$. According to statistical learning theory⁵⁴, the quality of the predictive surrogate can be quantified by measuring how closely its predictions match the true mean-value behavior. A standard metric for this purpose is the average prediction error over the data distribution $\mathbb{D}$, i.e.,

$$\mathbf{R}(\mathbf{h}) = \mathbb{E}_{\mathbf{x} \sim \mathbb{D}} |\mathbf{h}(\mathbf{x}, \mathbf{O}) - \mathbf{f}(\tilde{\rho}(\mathbf{x}), \mathbf{O})|^2. \quad (3)$$

The predictive surrogate $\mathbf{h}$ is considered computationally efficient if it can be trained in polynomial time with respect to the qubit count N and the input dimension d , and with high probability, satisfies $\mathbf{R}(\mathbf{h}) \leq \epsilon$.

Prior studies have shown that constructing efficient predictive surrogates for arbitrary quantum circuits $\mathbf{U}(\mathbf{x})$ is unlikely, as ruled out by complexity-theoretic constraints⁴⁴. Motivated by this, we pursue efficient predictive surrogate design under well-defined and practically relevant scenarios, and accordingly devise two provably efficient methods, denoted by h_{cs} and h_{qs} .

Predictive surrogate h_{cs}

The first predictive surrogate h_{cs} is designed for circuits $\mathbf{U}(\mathbf{x})$ with independently tunable RZ gates and supports arbitrary, yet K -local and

bounded, observables $\{\mathbf{O}\}$. The observable $\mathbf{O}$ is said to be K -local if it can be written as a sum of tensor products of Pauli operators, where each term acts nontrivially on at most K qubits and trivially on the rest. Additionally, the norm of the observable is bounded with $\|\mathbf{O}\|_\infty \leq B$. In this sense, h_{cs} functions as a classical shadow predictor, capable of estimating mean values of many local observables for previously unseen quantum states $\tilde{\rho}(\mathbf{x}')$ with $\mathbf{x}' \sim \mathbb{D}$, all without requiring further access to quantum processors.

The implementation of h_{cs} contains two steps: (i) data collection and (ii) model construction. In Step (i), the learner inputs various $\mathbf{x}^{(i)} \in [-\pi, \pi]^d$ to $\tilde{\mathcal{U}}(\mathbf{x}^{(i)})$ and collects classical information of the evolved state $\tilde{\rho}(\mathbf{x}^{(i)})$ under Pauli-based classical shadow⁵⁵ with T snapshots, denoted by $\tilde{\rho}_T(\mathbf{x}^{(i)})$. Following this routine, the training dataset $\mathcal{T}_{cs} = \{(\mathbf{x}^{(i)}, \tilde{\rho}_T(\mathbf{x}^{(i)}))\}_{i=1}^n$ with n examples is constructed. Then, in Step (ii), the learner utilizes $\mathcal{T}_{cs}$ to build the predictive surrogate. Given a new input $\mathbf{x}'$, the predicted classical shadow of the state $\rho(\mathbf{x}')$ yields

$$\hat{\sigma}_n(\mathbf{x}') = \frac{1}{n} \sum_{i=1}^n \kappa_\Lambda(\mathbf{x}', \mathbf{x}^{(i)}) \tilde{\rho}_T(\mathbf{x}^{(i)}), \quad (4)$$

where $\kappa_\Lambda(\mathbf{x}, \mathbf{x}^{(i)}) = \sum_{\omega, \|\omega\|_0 \leq \Lambda} 2^{\|\omega\|_0} \Phi_\omega(\mathbf{x}) \Phi_\omega(\mathbf{x}^{(i)})$ is the truncated trigonometric monomial kernel with the basis $\Phi_\omega(\mathbf{x}) = \prod_{l=1}^d [\mathbb{1}_{\omega_l=0} + \mathbb{1}_{\omega_l=1} \cdot \cos(\mathbf{x}) + \mathbb{1}_{\omega_l=-1} \cdot \sin(\mathbf{x})]$. Refer to Supplementary Information (SI) A for more details.

Supported by $\hat{\sigma}_n(\mathbf{x}')$, the predictive surrogate over the specified observable $\mathbf{O}$ is

$$\mathbf{h}_{cs}(\mathbf{x}', \mathbf{O}) = \frac{1}{n} \sum_{i=1}^n \kappa_\Lambda(\mathbf{x}', \mathbf{x}^{(i)}) \mathbf{g}(\mathbf{x}^{(i)}, \mathbf{O}), \quad (5)$$

where $\mathbf{g}(\mathbf{x}^{(i)}, \mathbf{O}) = \text{Tr}(\tilde{\rho}_T(\mathbf{x}^{(i)})\mathbf{O})$ refers to the shadow estimation of $\text{Tr}(\tilde{\rho}(\mathbf{x}^{(i)})\mathbf{O})$. The following theorem indicates that even in the noisy scenario, the proposed predictive surrogate can accurately approximate the mean value $\mathbf{f}(\tilde{\rho}(\mathbf{x}), \mathbf{O})$ in Eq. (2), where the formal statement and the proof are deferred to SI B.

Theorem 1. Following notations in Eqs. (1)–(5), denote $\mathcal{C}(\Lambda) = \{\omega | \omega \in \{0, \pm 1\}^d, \text{ s.t. } \|\omega\|_0 \leq \Lambda\}$ as the truncated frequency set with Λ being the threshold value. Suppose that the input $\mathbf{x} \sim \text{Unif}[-\pi, \pi]^d$ follows the uniform distribution and $\mathbb{E}_{\mathbf{x}} \|\nabla_{\mathbf{x}} \text{Tr}(\tilde{\rho}(\mathbf{x})\mathbf{O})\|_2^2 \leq C$. Then, under the Pauli noise channel $\mathcal{N}_p(p_X, p_Y, p_Z)$ with $p = \min\{p_X, p_Y\}$, when $\Lambda = 4C/\epsilon$ and the number of training examples is

$$n = \tilde{\Omega} \left(\left| \mathcal{C} \left(\min \left\{ \frac{4C}{\epsilon}, \frac{1}{2(p+p_Z)} \log \left(\frac{2B}{\sqrt{\epsilon}} \right) \right\} \right) \right| \frac{2B^2 9^k}{\epsilon} \right),$$

with probability at least $1 - \delta$, we have $\mathbf{R}(\mathbf{h}_{cs}) \leq \epsilon$ even for $T = 1$ snapshot per training example.

The results suggest that the sample complexity n does not explicitly depend on the qubit count N and polynomially scales with d for small C . Ref. 44 has empirically shown that the quantity C typically decreases with the increased N and the circuit depth. To ensure the non-triviality and scalability of h_{cs} , the gradient norm C must remain non-vanishing, thereby ruling out barren-plateau regimes that would otherwise cause h_{cs} to collapse to a trivial constant function. Refer to SI B for details.

In addition, the collected classical shadows $\{\tilde{\rho}_T(\mathbf{x}^{(i)})\}$ inherently encode the noise characteristics of the explored quantum processor. This allows h_{cs} to accurately emulate the mean values of noisy quantum states without explicitly characterizing the underlying Pauli noise. This distinguishes our approach from classical simulators^{29,56,57}, which require expensive computational overhead to achieve the same goal⁵³.

Moreover, different from the ideal case, the scaling of n is dominated by the noise parameters, as reflected in the term $\min\{4C/\epsilon, 1/(2(p+p_Z))\}$. As a result, higher levels of Pauli noise reduce

both $1/(2(p + p_z))$ and C^{58} , which in turn decreases the required n . This observation highlights the potential of h_{cs} for advancing the practical utility of near-term quantum devices.

Remark. The achieved results in Theorem 1 can be effectively extended to a broader class of quantum circuits with arbitrary Pauli rotation gates and Clifford gates, as explained in SI A. In SI E, we prove the computational efficiency of h_{cs} , scaling with $\mathcal{O}(n^2NT)$.

Predictive surrogate h_{qs}

The second predictive surrogate h_{qs} is tailored for $\mathbf{U}(\mathbf{x})$ with correlated parameters among RZ gates. This setting is commonly adopted in quantum simulations relevant to many-body physics⁴⁵ and quantum chemistry⁵⁹, as well as in quantum machine learning⁶⁰. Specifically, the quantum circuit in Eq. (1) yields a layer-wise layout, i.e., $\mathbf{U}(\mathbf{x}) = \prod_{j=1}^L (\prod_{i=1}^{d/L} \text{RP}(\mathbf{x}_j) \mathbf{V}_j)$, where RP refers to an arbitrary Pauli rotation gate, and the entries of $\mathbf{x} \in \mathbb{R}^d$ vary within each layer but remain the same across different layers. Without loss of generality, it is sufficient to set $\text{RP} = \text{RZ}$. The surrogate h_{qs} aims to predict the mean value $f(\tilde{\rho}(\mathbf{x}'), \mathbf{O})$ in Eq. (2) given a predefined observable $\mathbf{O}$ and any new input $\mathbf{x}'$.

The implementation of h_{qs} contains the same two steps as with h_{cs} , i.e., data collection and model construction, but differs in their execution. First, the learner builds the training dataset $\mathcal{T}_{qs} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^n$. The inputs $\mathbf{x}^{(i)}$ are sampled from an arbitrary distribution within the bounded interval $[-R, R]^d$, and $y^{(i)}$ is the estimated mean value of the observable $\mathbf{O}$ obtained by T shots. Here we suppose that the maximal estimation error satisfies $\max_{i \in [n]} |y^{(i)} - \text{Tr}(\tilde{\rho}(\mathbf{x}^{(i)})\mathbf{O})| \leq \epsilon_l$. Then, the predictive surrogate is defined by

$$h_{qs}(\mathbf{x}, \hat{\mathbf{w}}) = \langle \Phi_{\mathcal{E}(\Lambda)}(\mathbf{x}), \hat{\mathbf{w}} \rangle, \quad (6)$$

where $\hat{\mathbf{w}}$ is obtained by minimizing the loss function with $\min_{\mathbf{w}} \frac{1}{n} \sum_{i=1}^n (y^{(i)} - h_{qs}(\mathbf{x}^{(i)}, \mathbf{w}))^2 + \lambda \|\mathbf{w}\|_2$. The notation $\Phi_{\mathcal{E}(\Lambda)} = [\Phi_{\omega}(\mathbf{x})]_{\omega \in \mathcal{E}(\Lambda)}$ represents a feature vector with $|\mathcal{E}(\Lambda)|$ dimensions formed by the truncated trigonometric monomial $\Phi_{\omega}(\mathbf{x})$, and λ refers to the regularization parameter. For convenience, we denote the optimized surrogate $h_{qs}(\mathbf{x}, \hat{\mathbf{w}})$ as $h_{qs}(\mathbf{x})$ or h_{qs} , whenever no ambiguity arises. The subsequent theorem provides a provable guarantee for the efficiency of h_{qs} , where the formal description and proof are provided in SI C.

Theorem 2. Following notations in Eqs. (1)–(6), let $q = 1 - 2(p + p_z)$ and $\epsilon = 16B^2(\text{deg}(1+R)/\Lambda)^{2\Lambda}$. Suppose $q(1+R) < 1/e$ and $\epsilon_l \leq \sqrt{\epsilon}/4$. Then, when the frequency is truncated to $\Lambda > \text{deg}(1+R)$ and the number of training examples is

$$n = \left(\frac{1}{q(1+R)} \right)^{4\text{deg}(1+R)} \cdot \frac{\log(1/\delta)}{9},$$

with probability at least $1 - \delta$, we have $R(h_{qs}) \leq \epsilon$.

The achieved results indicate the capabilities of h_{qs} when applied to approximate $f(\tilde{\rho}(\mathbf{x}), \mathbf{O})$ with correlated $\mathbf{x}$. Concretely, the sample complexity and computational complexity of implementing h_{qs} scale polynomially with the input dimension d when the noise parameters satisfy $q \leq \mathcal{O}(1/d)$. This additional requirement compared to h_{cs} stems from the fact that the former considers more general settings. Different from h_{cs} designed for the uncorrelated $\mathbf{x}$ with a uniform input distribution, h_{qs} is tailored to accommodate the correlated $\mathbf{x}$ and arbitrary input distributions, making it applicable to a broader range of scenarios. This increased flexibility comes at the expense of involving additional conditions.

Remark. In SI D, we provide an alternative solution of h_{qs} , which can efficiently emulate the mean-value behavior of $f(\tilde{\rho}(\mathbf{x}), \mathbf{O})$ with low noise levels (i.e., $q > \mathcal{O}(1/d)$), provided that the range R is small.

Applications to digital quantum simulation

A crucial feature of predictive surrogates is that once trained, they can predict mean values during inference without any further access to the quantum processor. As a result, when the task is to estimate mean values across diverse $\{\rho(\mathbf{x})|\mathbf{x}\}$, the number of required quantum measurements, and thereby overall quantum processor usage, can be greatly reduced. This advantage is most evident in measurement-intensive applications. To illustrate this, we next present how h_{cs} and h_{qs} can be employed to advance two crucial tasks in digital quantum simulation, i.e., VQEs⁴⁸ and quantum phase identification, where more details are given in Methods and SI G.

VQEs leverage the variational principle to compute the ground state energy of quantum systems. Given a Hamiltonian $\mathbf{H}$, VQE adopts the parameter-shift rule⁶¹ to iteratively update $\mathbf{x}$ to minimize $f(\tilde{\rho}(\mathbf{x}), \mathbf{H})$ in Eq. (2), often incurring tremendous access to quantum processors. As indicated in ref. 44, a well-optimized $\mathbf{h} \in \{h_{cs}, h_{qs}\}$ can alleviate the expensive measurement overhead via pre-training. In particular, the appropriate choice of $\mathbf{h}$ is scenario dependent, determined by the adopted ansatz and the number of Hamiltonians considered (See SI F for details). Instead of directly optimizing $f(\tilde{\rho}(\mathbf{x}), \mathbf{H})$, the targeted parameters can be obtained by minimizing $h(\mathbf{x}, \mathbf{H})$, i.e.,

$$\min_{\mathbf{x}} h(\mathbf{x}, \mathbf{H}). \quad (7)$$

This process is entirely classical and does not involve any access to quantum processors. Given access to the optimized $\hat{\mathbf{x}}$, the unitary $\mathbf{U}(\hat{\mathbf{x}})$ can, if desired, be deployed on quantum processors for additional fine-tuning.

The predictive surrogate can also facilitate the identification of phase transitions in quantum many-body systems. A representative example is identifying Floquet symmetry-protected topological (FSPT) phases⁵⁰, which are exotic non-equilibrium states of matter arising in periodically driven systems⁶². Define a 1D spin-1/2 chain governed by the time-periodic Hamiltonian as

$$\mathbf{H}_{\text{tp}}(t) = \begin{cases} \mathbf{H}_{\text{tp},1} = (\frac{\pi}{2} - \delta) \sum_i \mathbf{X}_i, & 0 \leq t < T_1, \\ \mathbf{H}_{\text{tp},2} = -\sum_i J_i \mathbf{Z}_{i-1} \mathbf{X}_i \mathbf{Z}_{i+1}, & T_1 \leq t < 2T_1. \end{cases} \quad (8)$$

Here $\mathbf{P}_i \in \{\mathbf{X}, \mathbf{Y}, \mathbf{Z}\}$ refers to the Pauli operator on the i -th qubit. Through varying the drive perturbation δ , two dynamical phases emerge: FSPT phase and thermal phase. These two phases can be characterized by the local magnetization $\langle \mathbf{Z}_i(\mathbf{x}) \rangle = \langle 0^N | \mathbf{U}(\mathbf{x})^\dagger \mathbf{Z}_i \mathbf{U}(\mathbf{x}) | 0^N \rangle$, where the gate layout of $\mathbf{U}(\mathbf{x})$ depends on $\mathbf{H}_{\text{tp}}(t)$, and $\mathbf{x}$ is determined by the specified disordered coupling parameters $\{J_i\}$, t , and δ . Refer to SI H for more details about the circuit implementation. In this regard, identifying FSPT phases typically requires a large number of measurements to estimate $\langle \mathbf{Z}_i(\mathbf{x}) \rangle$ with varied $\mathbf{x}$. The proposed h_{qs} can effectively address this bottleneck.

Experimental results

We experimentally validate the efficacy of the proposed predictive surrogates in advancing digital quantum simulation. All experiments are carried out 42 active qubits within a 66-qubit programmable superconducting processor, which comprises a 6×11 array of frequency-tunable transmon qubits with tunable couplers. See Methods and SI H for a full device characterization and the omitted experimental results. For each predictive surrogate, we first assess its prediction accuracy on quantum processors and then demonstrate its effectiveness in the corresponding downstream tasks.

The first task is to explore the potential of h_{cs} to enhance VQE optimization in finding the ground state energy of 1D TFIMs. The formal expression is $\mathbf{H}_{\text{TFIM}} = -J \sum_i \mathbf{Z}_i \mathbf{Z}_{i+1} - h \sum_i \mathbf{X}_i$. The initial state is $\rho_0 = (|+\rangle\langle+|)^{\otimes N}$, and the adopted ansatz in VQE follows the Trotter

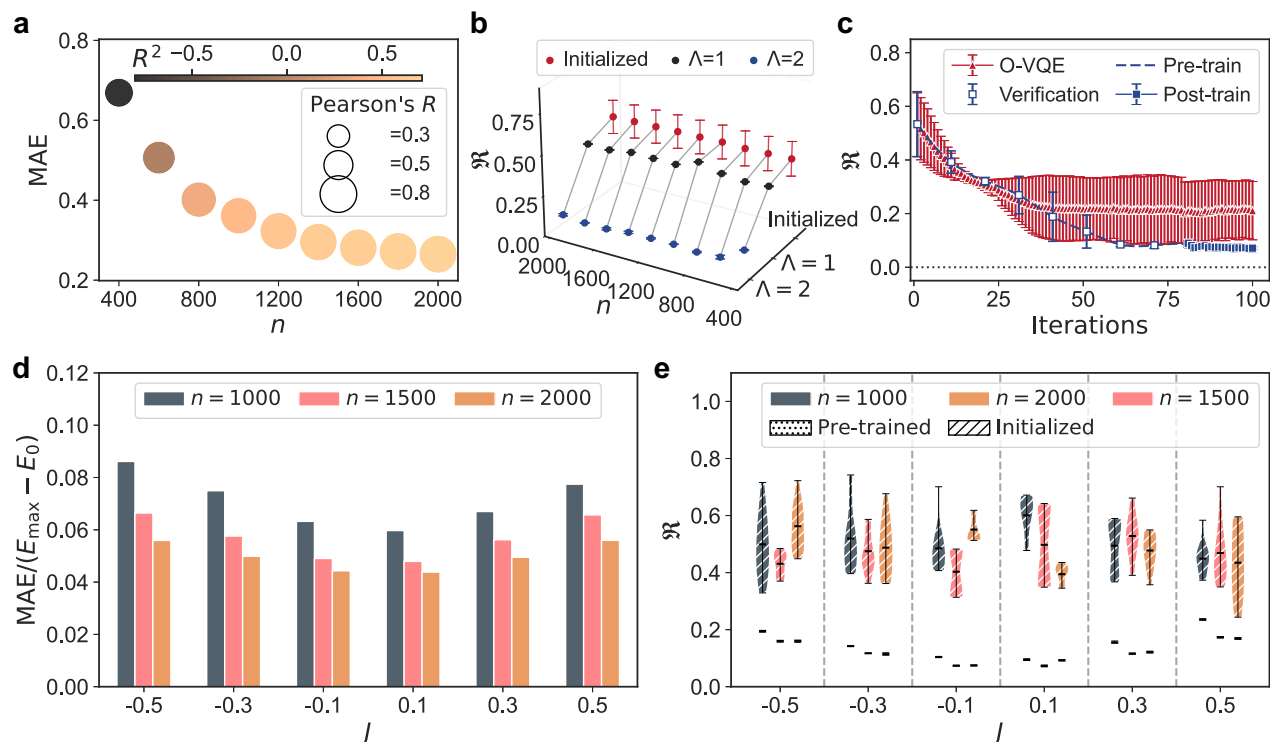


Fig. 2 | Experimental results of using the predictive surrogate h_{cs} to pre-train VQE for a family of 1D transverse field Ising models. **a** Quantifying the difference between the outputs of the employed quantum processor and the predictive surrogate in the measures of mean absolute error (MAE), coefficient of determination (R^2), and Pearson correlation coefficient (R). **b** Normalized deviation $\mathfrak{R}(\mathbf{x})$ of the estimated energy $f(\tilde{\rho}(\mathbf{x}), \mathbf{H}_{\text{TFIM}})$ from the ground state energy $E_0(\mathbf{H}_{\text{TFIM}})$, evaluated on both randomly chosen parameters (highlighted by red dots with the label 'Initialized') and parameters optimized by the predictive surrogate h_{cs} with varying the size of the training dataset n and the truncation threshold Λ (i.e., $\Lambda = 1, 2$ are highlighted by the black and blue dots, respectively). Error bars represent standard deviations over 20 independent instances. **c** Optimization dynamics of the normalized deviation $\mathfrak{R}(\mathbf{x})$ over 100 iterations for the predictive surrogate h_{cs} compared to the original VQE, corresponding to the labels of 'Pre-train' and 'O-VQE'. The blue solid box, labeled 'Post-train', refers to the optimization dynamic of the additional fine-tuning on the quantum processor. The blue hollow box, labeled 'Verification', indicates the mean values measured on the quantum processor at the

parameters optimized by the predictive surrogate. Error bars represent standard deviations over five independent experiments. **d** Normalized MAE between quantum processor outputs and predictions from three optimized h_{cs} , each trained on datasets of varied sizes (i.e., $n = 1000, 1500, 2000$), evaluated for a set of $\mathbf{H}_{\text{TFIM}}$ with varying $J \in \{\pm 0.1, \pm 0.3, \pm 0.5\}$ and fixed $h = -0.5$. **e** Normalized deviation $\mathfrak{R}(\mathbf{x})$ for randomly chosen parameters (highlighted by the hatched violin with the label 'Initialized') and parameters optimized by the predictive surrogate (highlighted by the dotted violin with the label 'Pre-trained') when J in $\mathbf{H}_{\text{TFIM}}$ ranges from -0.5 to 0.5 and fixed $h = -0.5$. The black horizontal line within each violin indicates the mean value. The vertical black lines extend from the minimum to the maximum values. Different colors correspond to different training dataset sizes ($n = 1000, 1500, 2000$). Due to the highly concentrated distribution of the pre-trained results (with a standard deviation orders of magnitude smaller than the y-axis scale), the violin bodies of the 'Pre-trained' appear as narrow lines. All the results presented in the subplot are statistical data collected from 20 independent experiments.

decomposition, i.e.,

$$\mathbf{U}(\mathbf{x}) = \prod_{l=1}^L \left(\prod_{i=1}^N \text{RX}_i(\mathbf{x}_{l,i}) \prod_{i=1}^{N-1} \text{RZZ}_{i,i+1}(\mathbf{x}_{l,N+i}) \right),$$

where the input dimension is $d = L(2N - 1)$. In what follows, we first evaluate the prediction accuracy of h_{cs} and then systematically examine its effectiveness for VQE pre-training, progressing from single- to multiple-Hamiltonian instances.

Our first subtask is to quantify the performance of h_{cs} in emulating the mean-value behavior of the VQE under study. The TFIM considered here is set to $N = 6, J = -0.1$, and $h = -0.5$. Moreover, the dataset $\mathcal{T}_{cs}$ is constructed by uniformly sampling $\mathbf{x}$, where the number of snapshots for each state $\rho(\mathbf{x})$ is fixed to be $T = 10$. The threshold of truncation is set to $\Lambda = 2$. The performance of $h_{cs}(\mathbf{x})$ over 200 test examples is shown in Fig. 2a. In particular, under the measures of the mean absolute error (MAE), Pearson correlation coefficient, and R^2 score, the optimized $h_{cs}(\mathbf{x})$ exhibits a strong agreement with $f(\tilde{\rho}(\mathbf{x}), \mathbf{H}_{\text{TFIM}})$ given $n \geq 1600$. These results not only validate the effectiveness of $h_{cs}(\mathbf{x})$, but also serve as a crucial prerequisite for successfully pre-training VQE.

We now proceed to examine the ability of the optimized h_{cs} in estimating the ground state energy of $\mathbf{H}_{\text{TFIM}}$. To this end, we use the

normalized deviation as a metric to quantify its performance, i.e.,

$$\mathfrak{R}(\mathbf{x}) = \frac{|f(\tilde{\rho}(\mathbf{x}), \mathbf{H}_{\text{TFIM}}) - E_0(\mathbf{H}_{\text{TFIM}})|}{E_{\text{max}}(\mathbf{H}_{\text{TFIM}}) - E_0(\mathbf{H}_{\text{TFIM}})},$$

where $\mathbf{x}$ refers either to the initialized inputs or the optimized parameters returned by h_{cs} . Here, $E_0(\mathbf{H}_{\text{TFIM}})$ and the dominator refer to the ground state energy and the spectrum width of $\mathbf{H}_{\text{TFIM}}$, respectively. We repeat the evaluation 20 times, each time using the same set of initialized parameters, followed by the parameter optimization described in Eq. (7). As depicted in Fig. 2b, in every case, the average $\mathfrak{R}(\mathbf{x})$ before pre-training remains near 0.482, regardless of n used to implement h_{cs} . In addition, when $\Lambda = 1$, the estimated $f(\tilde{\rho}(\mathbf{x}), \mathbf{H}_{\text{TFIM}})$ with $\tilde{\mathbf{x}}$ being return by pre-training h_{cs} still suffers from a high $\mathfrak{R}(\tilde{\mathbf{x}})$ with all above 0.42. In contrast, when $\Lambda = 2$ and $n > 600$, $\mathfrak{R}(\tilde{\mathbf{x}})$ is less than 0.09 after pre-training. These results hint that when Λ surpasses a critical value, its performance improves with a large sampling size n , aligning with theoretical results in Theorem 1.

To further comprehend the potential of pre-training VQE, we compare the optimization dynamics of the predictive surrogate with those of the original VQE. Specifically, the explored $\mathbf{H}_{\text{TFIM}}$ in this analysis is identical to that employed in the previous subtask, and the

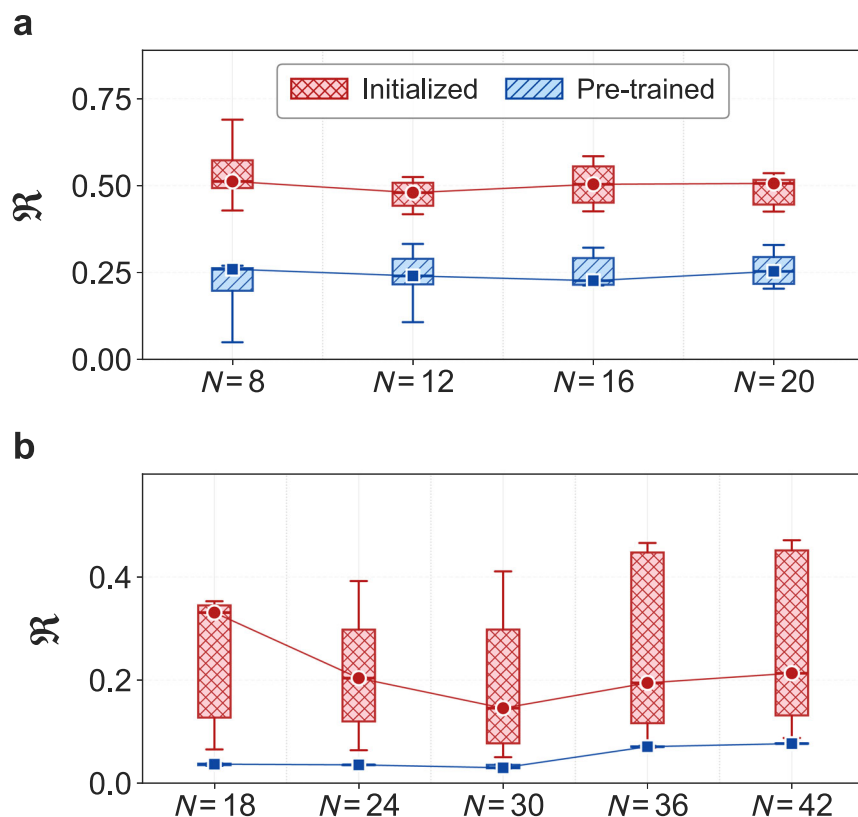


Fig. 3 | Scaling behavior of h_{cs} and h_{qs} when applied to pre-train VQE.

a Normalized deviation $\mathcal{R}(\mathbf{x})$ for h_{cs} evaluated on the initial parameters before pre-training (highlighted by red box with the label 'Initialized') and after pre-training (highlighted by blue box with the label 'Pre-trained') across varying qubit counts $N \in \{8, 12, 16, 20\}$ and fixed training data size $n = 1024$. **b** Normalized deviation $\mathcal{R}(\mathbf{x})$

for h_{qs} evaluated on the initial parameters before and after pre-training (with labels following the same convention in the upper panel) across varying $N \in \{18, 24, 30, 36, 42\}$ and fixed $n = 160$. The results are aggregated over 20 random instances per experimental setting.

hyperparameter settings of h_{cs} are $n = 2000$, $T = 10$, and $\Lambda = 2$. The learning rate is set as 0.1, and the total number of iterations employed in pre-training VQE and the original VQE is fixed to be 100. We repeat each setting 5 times to acquire the statistical results. As shown in Fig. 2c, after the pre-training phase, the VQE already outperforms the original VQE, yielding $\mathcal{R} = 0.09$ compared to 0.21. Remarkably, this improvement is attained while using only 0.023% of the quantum measurement resources required by the original VQE (Refer to SI H for details). More interestingly, the obtained $\mathcal{R}$ during the pre-training stage is near optimal, while only marginal gains are observed when further fine-tuning $\hat{\mathbf{x}}$ on the quantum processor, improving $\mathcal{R}$ from 0.09 to 0.07.

To assess the performance of the optimized h_{cs} in estimating the ground state energies of many $\mathbf{H}_{TFIM}$ without any access to the quantum processor, we undertake a new subtask. The TFIMs considered here span six different settings with $J \in \{-0.5, -0.3, -0.1, 0.1, 0.3, 0.5\}$, with h being fixed at -0.1 . The predictive surrogates with $\Lambda = 2$ are separately optimized using three different training set sizes, i.e., $n \in \{1000, 1500, 2000\}$. All other hyperparameter settings remain identical to those used in previous subtasks. For each optimized predictive surrogate, the pre-training stage is terminated upon convergence. Experimental results are visualized in Fig. 2e. For all TFIMs, the optimized surrogates achieve a lower $\mathcal{R}$ after pre-training compared to their initial values. Furthermore, the surrogates trained with $n = 1500$ and $n = 2000$ exhibit comparable performance, suggesting diminishing returns beyond a certain training set size. These findings confirm the capability of h_{cs} in substantially reducing the measurement overhead for VQE optimization, especially as the number of Hamiltonians grows.

We last explore the scalability of both predictive surrogates h_{cs} and h_{qs} for pre-training VQE. In particular, for h_{cs} we consider $\mathbf{H}_{TFIM}$ with varying numbers of qubits $N \in \{8, 12, 16, 20\}$ and a fixed number of training examples $n = 1024$. For h_{qs} , we consider the 2D-TFIMs $\mathbf{H}_{2D-TFIM}$ defined on square-lattice patches with $N \in \{18, 24, 30, 36, 42\}$ and $n = 160$. For all instances, the coefficients J and h are fixed at -0.1 and -0.5 . Refer to SI H for more details. As shown in Fig. 3 (i.e., the center line of each box indicates the median, the box limits represent the upper and lower quartiles, the whiskers correspond to the 10th and 90th percentiles), the results show that both h_{cs} and h_{qs} maintain comparable prediction performance as the system sizes N increase when the number of training examples n is fixed, and both consistently reduce the values of $\mathcal{R}(\mathbf{x})$ after pre-training. These results validate the effectiveness of the two surrogates across the considered range of N , and indicate that the surrogates' performance is largely independent of the number of qubits N , which aligns with Theorems 1 and 2, demonstrating their scalability.

We next turn to the second task, which is employing the predictive surrogate h_{qs} to identify FSPT phases and thermalized phases in a class of 20-qubit time-periodic Hamiltonians. Mathematically, this class of Hamiltonians is defined with 40 driving periods that increase monotonically, i.e., $\{\mathbf{H}_{tp}(t)\}$ with $t \in [kT_1]$, $k \in [79]$, and $2T_1$ being the driving period. This amounts to evaluating the mean value $\langle \mathbf{Z}_i(\mathbf{x}) \rangle$ with diverse $\mathbf{x}$. By convention, the values of $\mathbf{x}$ are generated by extensive sampling of $\{J\}$ and δ within the ranges $[0, 1]$ and $[0, 2]$, respectively.

To reduce the measurement overhead, we implement 20×79 predictive surrogates $\{h_{qs}\}$, each corresponding to a circuit configuration $\mathbf{U}(\mathbf{x})$ with a given driving time kT_1 and a qubit index i to be measured. For each h_{qs} , the size of $\mathcal{T}_{qs}$ is set to $n = 800$ and the number

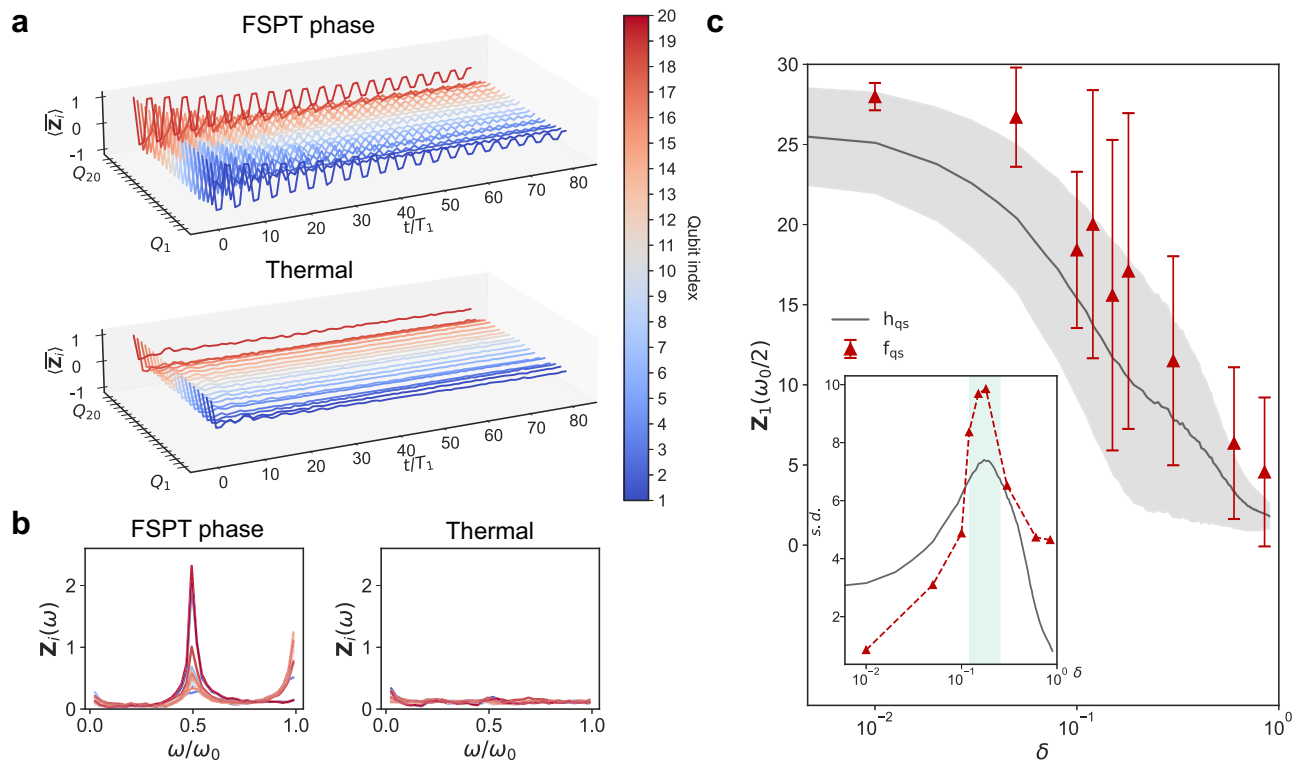


Fig. 4 | Experimental results of using the predictive surrogate h_{qs} to characterize the phase diagram of a 20-qubit 1D time-periodic Hamiltonian. **a** Time evolution of disorder-averaged local magnetizations $\{\langle \bar{Z}_i \rangle\}_{i=1}^{20}$ predicted by h_{qs} in the FSPT phase ($\delta = 0.01$) and the thermal phase ($\delta = 0.8$). A stable subharmonic oscillation of local magnetizations that persists over 40 driving periods could only be observed at the bound qubits (i.e., Q_1 and Q_{20}) in the FSPT phase, while magnetizations at bulk qubits (i.e., from Q_2 to Q_{19}) in the FSPT phase and all magnetizations in the thermal phase decay quickly to zero. **b** Fourier transform of local magnetization $\langle Z_i(\mathbf{x}) \rangle$ predicted by h_{qs} in the FSPT and thermal phases. The label of the x axis, ω/ω_0 , refers to the normalized frequency with $\omega_0 = \pi/T_1$ being the driven

frequency. The boundary qubits (spins) lock to the subharmonic frequency, which is in sharp contrast to the bulk spins. **c** The subharmonic peak height, i.e., $Z_1(\omega_0/2)$ with respect to the varied δ from 0 to 1. The results are obtained from the employed quantum processor (labeled by ' f_{qs} ') and the predictive surrogate (labeled by ' h_{qs} '). The data points shown for each δ are averaged over 5000 random instances of coupling parameters $\{J_i\}$ for the predictive surrogate and over 50 random instances for the adopted quantum processor. The shaded regions and error bars represent the standard error over these random instances, respectively. Inset: the standard deviation of the central peak height as a function of δ . The green shaded region refers to the identified region where the phase transition occurs.

of shots to obtain $y^{(i)}$ is $T = 10,000$. Refer to SI H for the satisfactory performance of the optimized predictive surrogates in the measure of $R(h_{qs})$ in Eq. (3).

Once all predictive surrogates are optimized, we use them to identify the phase transition point entirely on the classical side. In particular, for each surrogate h_{qs} , we consider two values of δ , which are 0.01 (FSPT phase) and 0.8 (thermal phase). For each δ , the remaining entries in $\mathbf{x}$, i.e., disordered coupling parameters $\{J_i\}$, are randomly sampled from a predefined distribution. This procedure is repeated 20 times, and 20×79 local magnetizations predicted by each surrogate are averaged over these trials, denoted by $\{\langle \bar{Z}_i(\mathbf{x}) \rangle\}$. As shown in Fig. 4a, the optimized surrogates successfully capture the signatures of distinct quantum phases. For a small perturbation $\delta = 0.01$, the values of $\{\langle \bar{Z}_i(\mathbf{x}) \rangle\}$ exhibit stable subharmonic oscillations at boundary spins (i.e., Q_1 and Q_{20}) over 40 driving periods, while decaying quickly to zero in the bulk of the chain (i.e., the qubit indices are from Q_2 to Q_{19}). In contrast, for a large perturbation $\delta = 0.8$, the magnetizations at both the edge and bulk spins decay quickly to zero within a few periods, signaling the thermalized phase.

We take a step further by quantifying the differences between these two phases through their spectral signatures. To this end, we apply the Fourier transform to the average local magnetization predicted by 20×79 predictive surrogates. As visualized in Fig. 4b, for the FSPT phase with $\delta = 0.01$, the Fourier spectra of $\{\langle \bar{Z}_i(\mathbf{x}) \rangle\}$ reveal that the boundary spins lock to the

subharmonic driven frequency $\omega/\omega_0 = 0.5$ with $\omega_0 = \pi/T_1$, whereas the bulk spins do not, showing a distinct difference between edge and bulk behavior. In contrast, in the thermal phase with $\delta = 0.8$, neither the edge spins nor the bulk spins exhibit a pronounced subharmonic frequency peak.

We last exploit the ability of the optimized $\{h_{qs}\}$ to identify the critical value δ^* , at which the phase transition occurs. According to the setting in ref. 50, the critical point amounts to the largest variance of the subharmonic spectral peak height. This peak height is defined as the amplitude in the Fourier spectrum of the local magnetization $\langle Z_i(\mathbf{x}) \rangle$ measured at $\omega = \omega_0/2$ for the boundary spin. Figure 4c presents predicted subharmonic peak height as a function of δ , obtained from h_{qs} over 100×5000 random parameters $\{(\delta, J_i)\}$. The inset of Fig. 4c shows a variance peak at $\delta = 0.172$. Using the 90% peak height width of the surrogate's predictions as a criterion, this indicates the phase transition occurs near this value. Experimental validation using quantum processors on a smaller dataset of 9×50 random parameters $\{(\delta, J_i)\}$ confirms the probable transition region. All of these results are consistent with the expected signature of FSPT and thermal phases as experimentally demonstrated in ref. 50, confirming the efficacy of h_{qs} to characterize non-equilibrium phase transitions.

Discussion

In this study, we propose two predictive surrogates designed to emulate the mean-value behavior of large-scale quantum processors. Theoretically, we prove the computational efficiency of these two

predictive surrogates in practical scenarios. Moreover, we experimentally demonstrate the capabilities of these predictive surrogates in advancing digital quantum simulation. Experimental results in pre-training VQEs and the identification of the FSPT phase up to 20 superconducting qubits indicate that our approaches not only substantially reduce measurement overhead compared to conventional quantum algorithms, but may also surpass their performance.

Our study opens a new research direction by harnessing advanced AI techniques to accelerate the pursuit of practical quantum utility. Several important research avenues remain to be explored. A promising avenue is to investigate how predictive surrogates can advance additional variational quantum algorithms^{63,64} and classical shadow estimation tasks⁴⁶. Next, it is important to generalize the achieved results to the circuits with intrinsic symmetries. This motivation is twofold: many quantum algorithms leverage the underlying symmetries of the problem Hamiltonian to reduce gate counts and enhance performance^{65–67}; and it is intriguing to investigate whether incorporating symmetry into quantum circuits can further improve the computational efficiency of predictive surrogates. Another important research direction is to extend the emulation of mean-value behavior to scenarios involving the learning of nonlinear functions, e.g., entanglement entropy, stabilizer entropy, and purity. Any progress in this direction would benefit a range of fields, such as quantum many-body physics^{68–70} and quantum system certification^{71–74}. In addition, it is important to investigate the development of predictive surrogates that can emulate output distributions rather than just mean values, as many quantum algorithms with provable advantages rely on sampling from the evolved state instead of estimating expectation values^{75–77}. Advancements in this direction would further broaden the potential applications of predictive surrogates. Parallel to research on provably efficient predictive surrogates, an important complementary direction is to systematically investigate whether deep learning-based surrogates^{78–80} can further enhance the efficiency in emulating quantum processors' behavior at scale.

Methods

Algorithmic framework for pre-training VQEs

The algorithmic implementation of pre-training VQEs with h_{cs} involves three steps:

1. Construction of predictive surrogates h_{cs} ;
2. Pre-training VQEs with h_{cs} on classical processors;
3. Optional fine-tuning with $f(\tilde{\rho}(\mathbf{x})\mathbf{H})$ on quantum processors. In the following, we briefly introduce the implementation of each step. Refer to SI G for more details.

Step 1. Given the training dataset $\mathcal{T}_{cs}$, the predictive surrogate takes the form of $h_{cs}(\mathbf{x}, \mathbf{H}; \mathbf{w})$ and is constructed and optimized.

Step 2. Given access to the optimized predictive surrogate h_{cs} , the optimal parameters $\mathbf{x}^*$ for VQE that minimize $f(\tilde{\rho}(\mathbf{x})\mathbf{H})$ is approximated by a set of parameters $\hat{\mathbf{x}}$ that minimize $h_{cs}(\mathbf{x}, \mathbf{H})$, i.e.,

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} h_{cs}(\mathbf{x}, \mathbf{H}).$$

The optimization is performed exclusively on the classical side, enabling the automatic differentiation with efficient classical optimizers. This suggests a substantial reduction of measurement cost, as conventional VQEs rely on the parameter-shift rule to perform optimization directly on quantum processors.

Step 3. After optimization, the quantum circuit $\mathbf{U}(\hat{\mathbf{x}})$ can be further optionally trained on the employed quantum processor by minimizing $f(\tilde{\rho}(\hat{\mathbf{x}}), \mathbf{H})$ via the parameter-shift rule.

Implementation of the FSPT phase identification

As explained in the main text, identifying FSPT phases requires estimating a set of expectation values $\langle \mathbf{Z}_i(\delta, \mathbf{J}, t) \rangle$ for varied observables $\{\mathbf{Z}_i\}_{i=1}^n$ and ansätze $\mathbf{U}(\delta, \mathbf{J}, t)$ with $t \in \{kT_1\}_{k=1}^{n_k}$. To this end, Nn_k

predictive surrogates $\{h_{qs}^{(i,t)}\}$ are separately constructed to emulate mean-value behaviors of $\langle \mathbf{Z}_i(\delta, \mathbf{J}, t) \rangle$. In the following, we briefly introduce the employed quantum circuits and the construction rule of h_{qs} . More details are deferred to SI G.

Let parameters $\mathbf{x}$ denote the triplet $(\delta, \mathbf{J}, t)$. Following the conventions of ref. 50, the circuit $\mathbf{U}(\mathbf{x})$ takes the form as

$$\mathbf{U}(\mathbf{x}) = (\mathbf{U}_2(\mathbf{J})\mathbf{U}_1(\delta))^{t/2T_1},$$

where $\mathbf{U}_1(\delta) = \exp(-iH_{tp,1})$, $\mathbf{U}_2(\mathbf{J}) = \exp(-iH_{tp,2})$, and $t/2T_1$ refers to the evolution periods. As such, the evolution time t determines the circuit depth and the number of correlated parameters $\mathbf{x}$.

The process of constructing the predictive surrogates $\{h_{qs}^{(i,t)}\}$ with varied observable $\mathbf{Z}_i$ and ansatz $\mathbf{U}(\mathbf{x})$ is the same. In particular, given access to the training dataset $\mathcal{T}_{qs}$ collected from the quantum processors for $\mathbf{Z}_i$ and $\mathbf{U}(\mathbf{x})$, the predictive surrogate follows the same form as defined in Eq. (6) with the feature map replaced by a subvector $\Phi_{Q(\Lambda)}(\mathbf{x})$ of the truncated feature vector $\Phi_{\mathcal{C}(\Lambda)}$, namely $h_{qs}^{(i,t)}(\mathbf{x}, \hat{\mathbf{w}}) = \langle \Phi_{Q(\Lambda)}(\mathbf{x}), \hat{\mathbf{w}} \rangle$. Here, $\Phi_{Q(\Lambda)}(\mathbf{x})$ is constructed by randomly and uniformly sampling $|Q(\Lambda)|$ elements without replacement from the set of trigonometric monomials $\{\Phi_{\omega}(\mathbf{x}) | \omega \in \mathcal{C}(\Lambda)\}$. This modification aims to balance the computational cost of $\Phi_{\mathcal{C}(\Lambda)}(\mathbf{x})$ and the error introduced by $\Phi_{\mathcal{C}(\Lambda-1)}(\mathbf{x})$.

Experiment setups

Here, we provide a brief overview of the device information for the quantum and classical processors used in the experiments, as well as the hyperparameter settings omitted in the main text. Refer to SI H for more details.

All experiments are conducted on a programmable superconducting processor. The selected qubits exhibit coherence times with a median $T_1 = 32.3 \mu\text{s}$, while achieving median gate errors of 0.11% for single-qubit operations and 1.09% for CZ gates.

All classical computations are performed on a computing server equipped with an NVIDIA A100 GPU (80GB HBM2 memory), accelerated via CUDA 12.2. The host system features dual Intel Xeon Gold 6348 CPUs (2.60GHz, 28 cores/socket).

Calculations leverage the JAX framework (v0.4.30) for hardware-accelerated computation. FP64 is used for surrogate model construction and prediction, while complex128 data types are employed for numerical quantum simulations.

For the surrogate models, the regularization parameter is set to $\lambda = 1$, and the feature map $\Phi_{Q(\Lambda)}$ is constructed by randomly sampling 1000 features from $\mathcal{C}(\Lambda)$ with the frequency threshold fixed at $\Lambda = 2$. For the 1D TFIM experiments (using h_{cs}) with system sizes $N \in \{8, 12, 16, 20\}$, the quantum circuit depth is $L = 1$, and the input parameters $\mathbf{x}$ are uniformly sampled from the interval $[-\pi, \pi]^d$ with $d = L(2N - 1)$. For the 2D TFIM experiments (using h_{qs}) with $N \in \{18, 24, 30, 36, 42\}$, the circuit depth is also $L = 1$, and the parameters $\mathbf{x}$ are uniformly sampled from $[-\pi, \pi]^d$ with $d = 2L$. In all cases, we use the ADAM optimizer with a learning rate of $\eta = 0.1$ for pre-training the VQEs and for additional fine-tuning on quantum processors⁸¹.

The training parameters δ and the disorder coupling parameters $\mathbf{J}$ are independently sampled from the uniform distributions over $(0, 1)$ and $[0, 2]^N$, respectively.

When constructing the predictive surrogates h_{qs} , we set the regularization parameter to $\lambda = 1$ and construct the feature map $\Phi_{Q(\Lambda)}$ by randomly sampling 1000 features from $\mathcal{C}(\Lambda)$ with the frequency threshold set to $\Lambda = 25$.

Data availability

The data generated and analyzed during the current study are publicly available at the following repositories: Github: https://github.com/weiyouliao/Demonstration_of_Efficient_Predictive_Surrogates_for_

Large-scale Quantum Processors; OSF: https://osf.io/dgyzf/overview?view_only=9248516af81c40d2a3aa5e899e7c34e0.

Code availability

The code used in this study is available at the GitHub repository https://github.com/weiyouliao/Demonstration_of_Efficient_Predictive_Surrogates_for_Large-scale_Quantum_Processors.

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Author contributions

The project was conceived by Y.D., D.T. and H.H. Theoretical results were proved by Y.D. and X.W. Experiments and results analysis were performed by W.L., Y.D., X.W., T.T. and H.H. All authors contributed to the write-up.

Competing interests

The authors declare no competing interests.

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